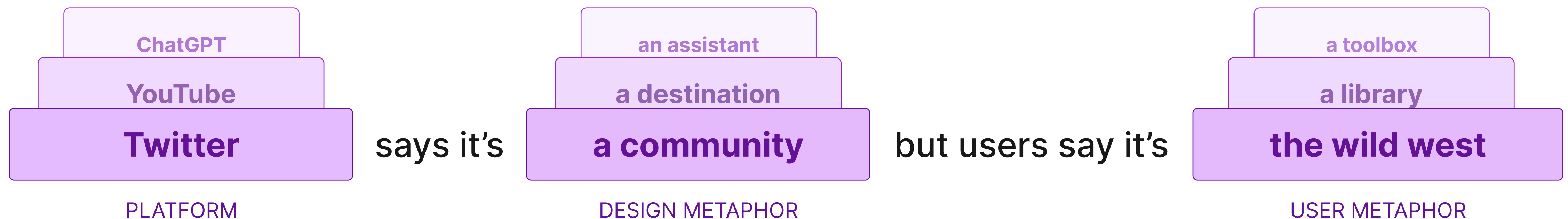


Comparing Design Metaphors and User-Driven Metaphors

Beleicia Bullock, James A. Landay, Michael S. Bernstein



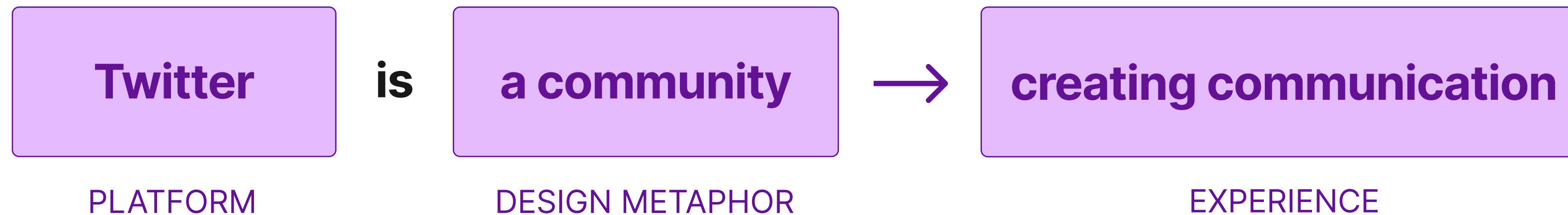
Interactive systems *rely* on design metaphors to shape ideal user experiences.



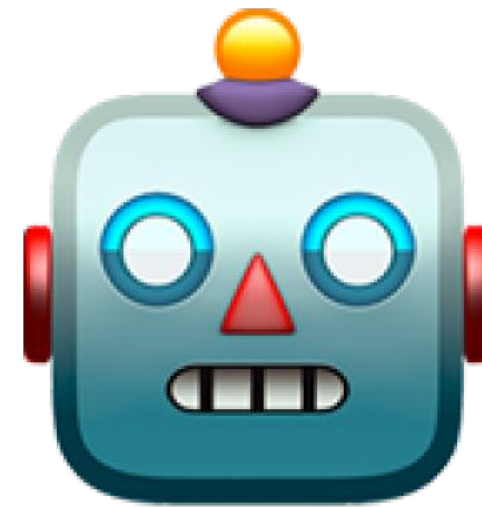
Design metaphors shape and communicate the experiences *we want* users to have.



Design metaphors shape and communicate the experiences *we want* users to have.



Design metaphors shape and communicate the experiences *we want* users to have.



Prior work has explored ways to generate “good” design metaphors...

Carroll et al. (1988), Anderson et al. (1994), Logler et al. (2018), Lockton et al. (2019), Cila et al. (2012), Cila et al. (2014)



guidelines and theories

what is considered a “good” design metaphor?



tools

how do we make design metaphor generation easier?



empirical studies

how do metaphors differ across different generation strategies?

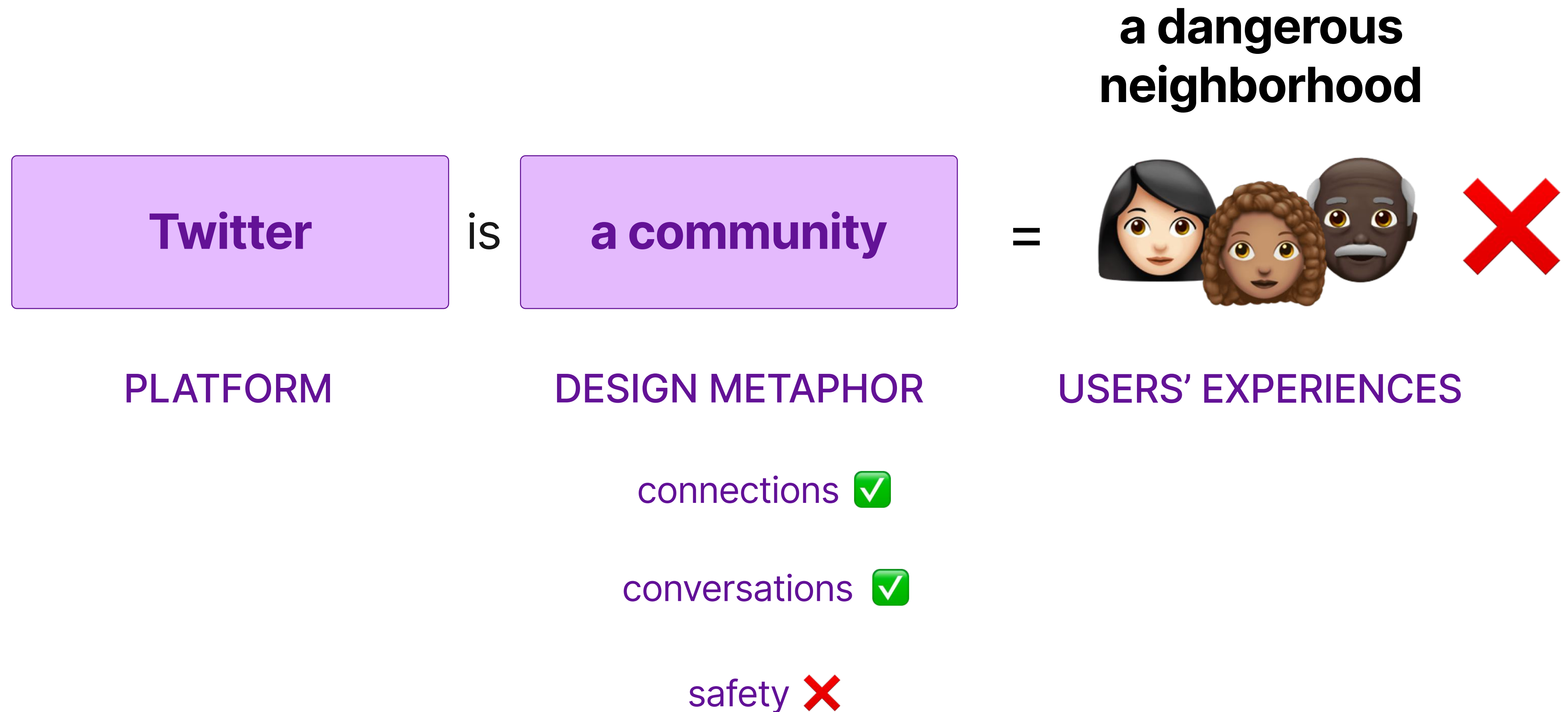
.....but we don't know if users experience our “good” design metaphors.



What if users do not experience our design metaphors?



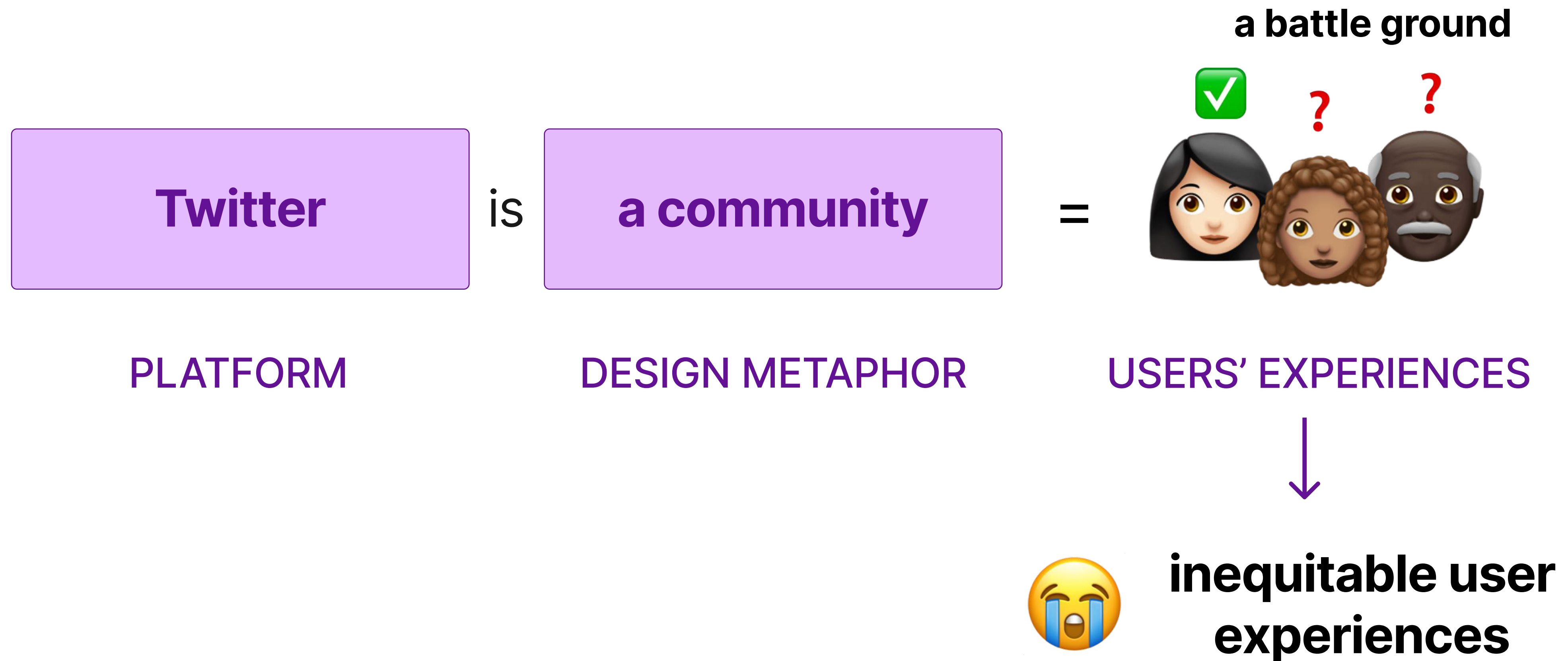
What if users do not experience our design metaphors?



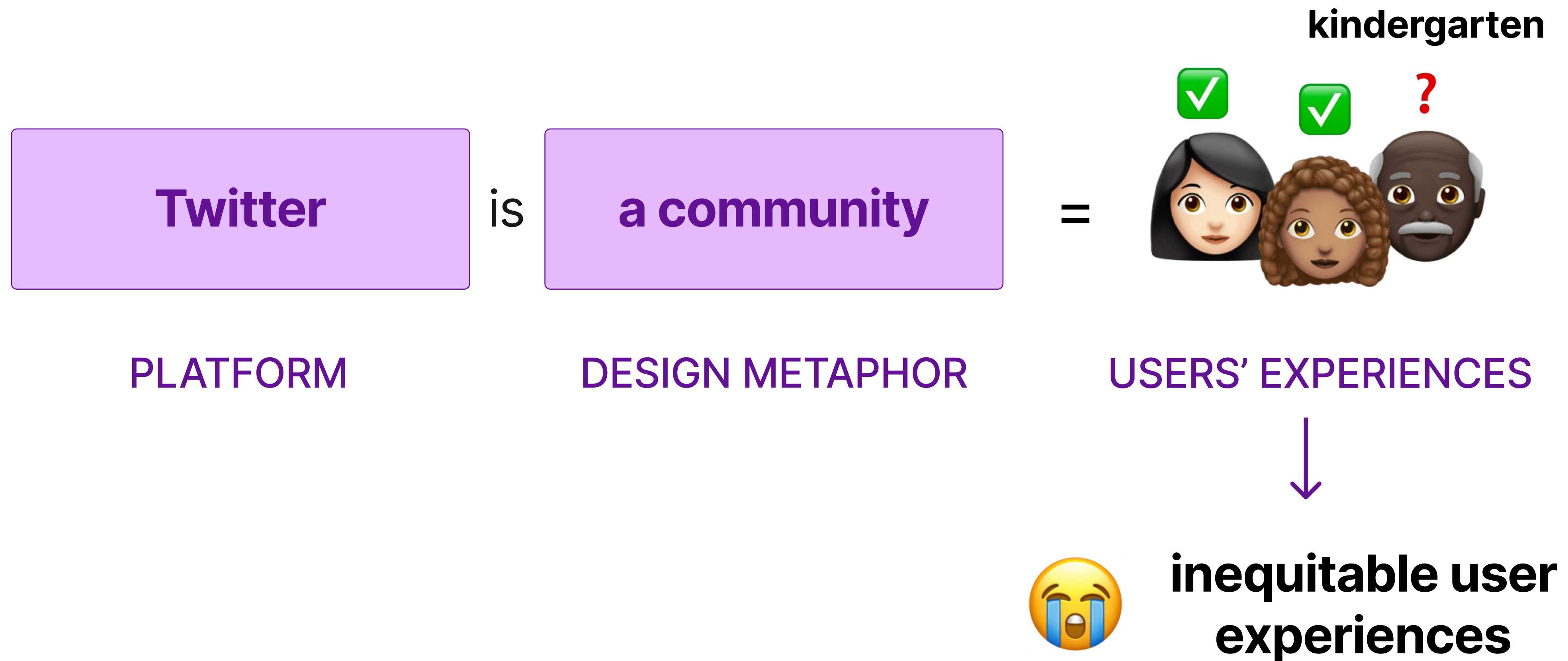
If our design metaphors do translate, do they translate to *all* users or just some?



If our design metaphors do translate, do they translate to *all* users or just some?



If our design metaphors do translate, do they translate to *all* users or just some?



We propose that “good” metaphors need to be evaluated by *comparing* design metaphors with users’ *experienced* metaphors.

Developing a method for comparing design and user metaphors



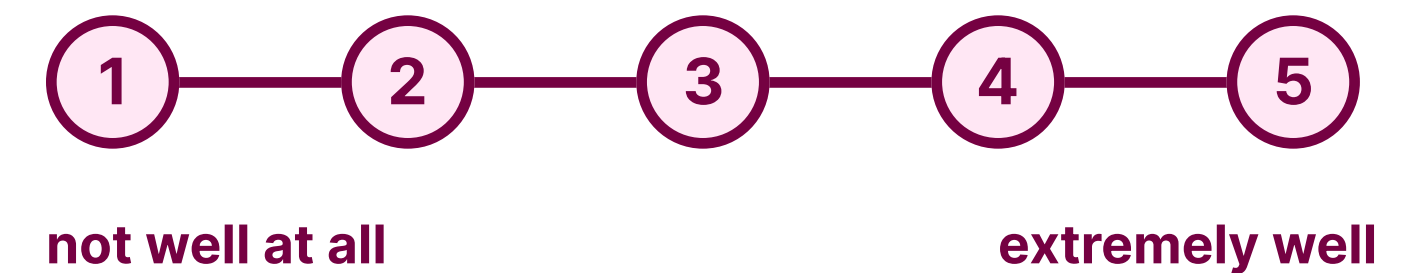
identifying design metaphors

what metaphors do platforms project to users?

Twitter is
because

eliciting user metaphors

what metaphors best describe users' experiences?

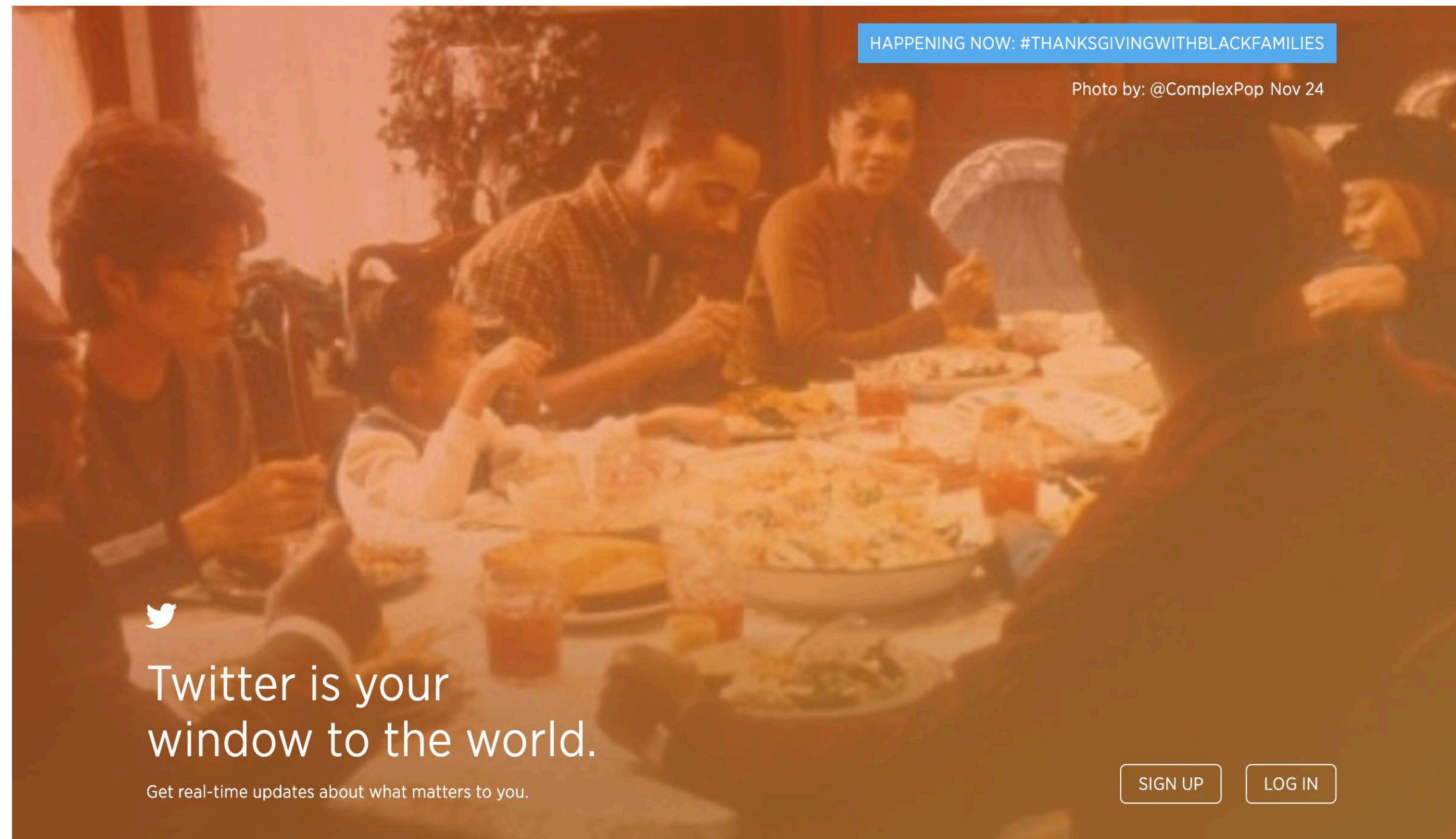


rating user metaphors

how well does a user metaphor reflect others' experiences?

Surfacing design metaphors via discourse analysis

Hoffman et al. (2018), Christin et al. (2024)

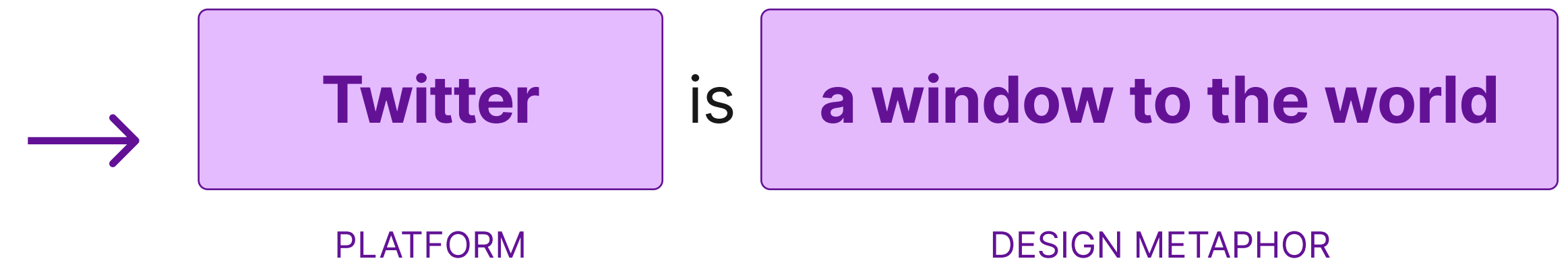
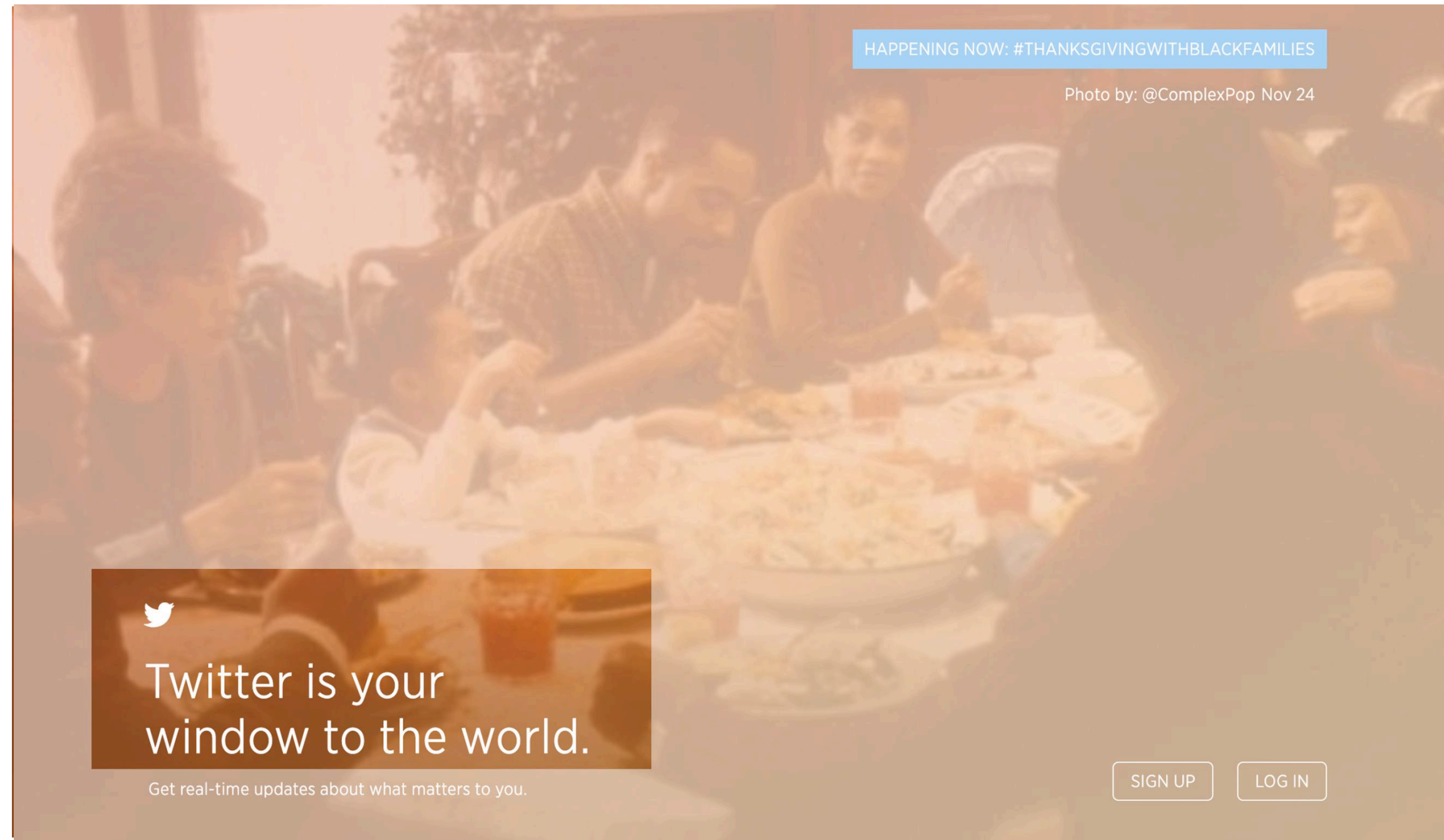


webpages as discourse

utilizing the home and about pages as places for insights about the platform

Surfacing design metaphors via discourse analysis

Hoffman et al. (2018), Christin et al. (2024)



webpages as discourse

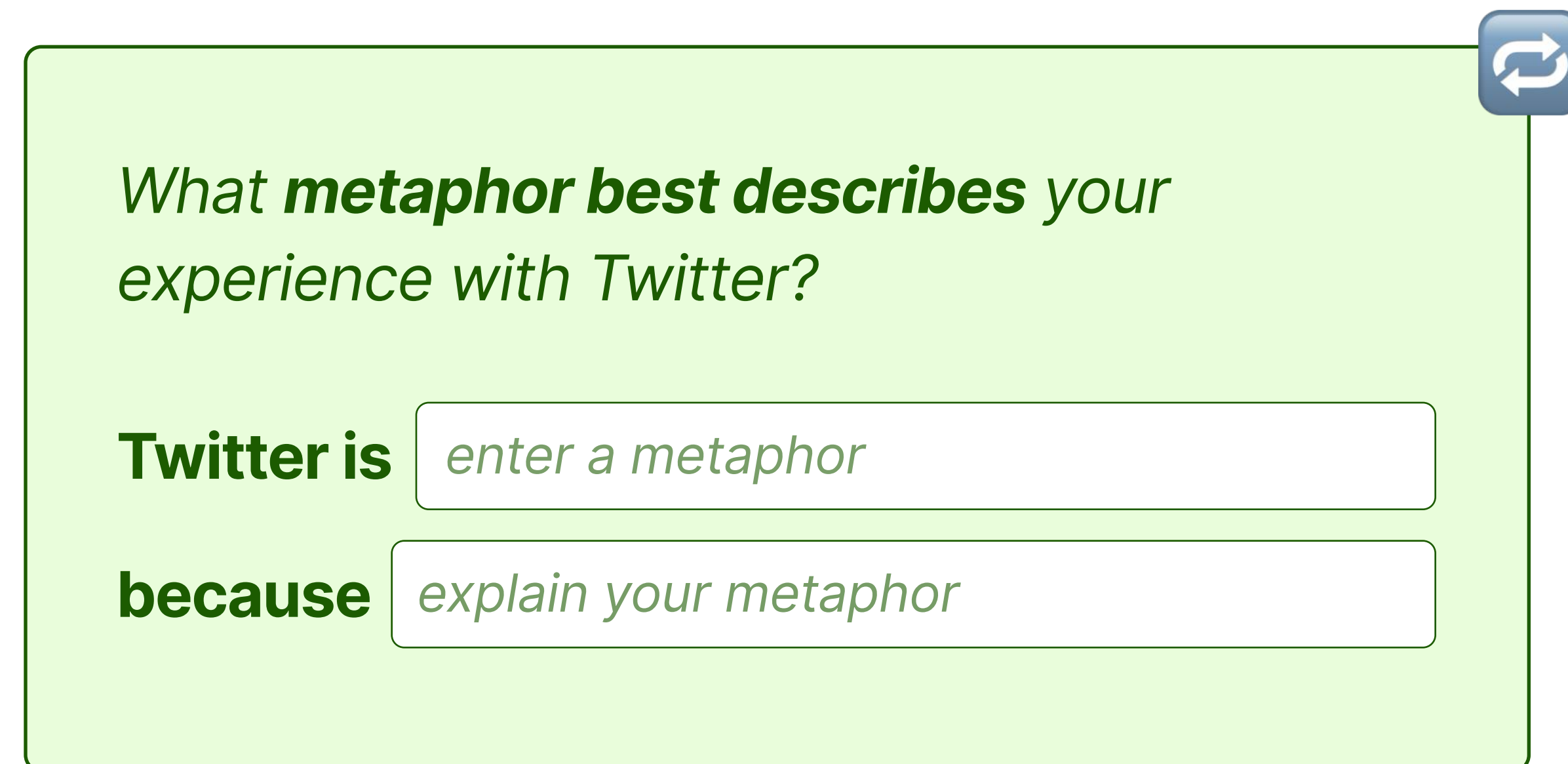
utilizing the home and about pages as places for insights about the platform

identifying platform definitions

generating design metaphors using “[PLATFORM] is...” statements

Eliciting user metaphors via sentence completion

French et al. (2017), Cheng et al. (2025), Lallemand and Mercier (2022), Schmidt et al. (2018)



*What **metaphor best describes** your experience with Twitter?*

Twitter is

because

sentence completion for user metaphors

conducting two elicitation rounds to surface metaphors that are unique and reflective to the user

Eliciting user metaphors via sentence completion



What **metaphor best describes** your experience with Twitter?

Twitter is

because

→

Twitter is **a jungle** because there are great, valuable things you can find in it, but it's also dangerous

Twitter is **the newsroom** because you can hear about all the latest news and developments on there

sentence completion for user metaphors

conducting two elicitation rounds to surface metaphors that are unique and reflective to the user

Rating user metaphors to understand broader experiences

How well does the following **metaphor reflect your experience** with Twitter?

*Twitter is **a jungle** because there are great, valuable things you can find in it, but it's also dangerous.*



rating user metaphors to understand saliency

a larger set of participants rate user metaphors based on how well they reflect their own experiences

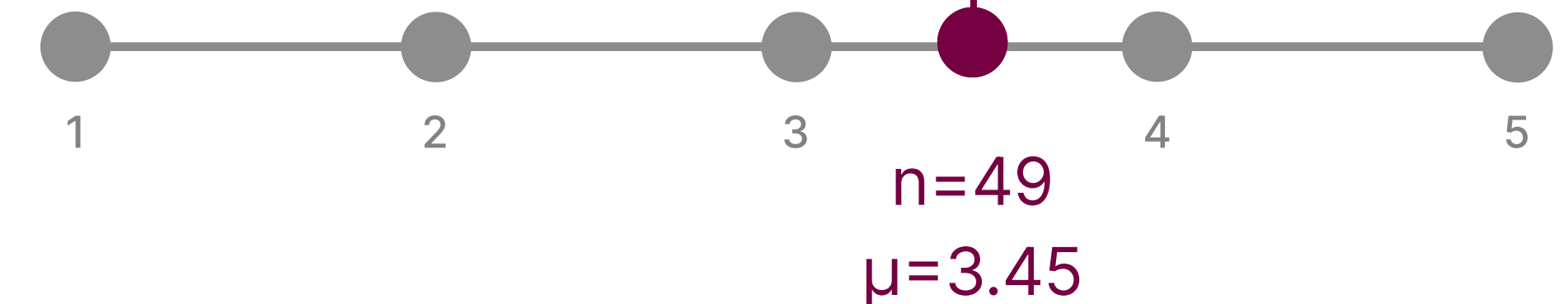
Rating user metaphors to understand broader experiences

How well does the following **metaphor** reflect your **experience** with Twitter?

Twitter is **a jungle** because there are great, valuable things you can find in it, but it's also dangerous.



Twitter is **a jungle** because there are great, valuable things you can find in it, but it's also dangerous.



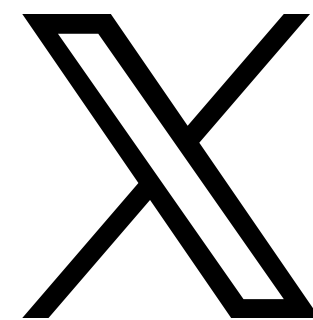
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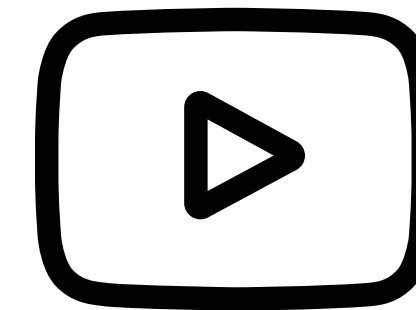
Deploying a method for comparing design and user metaphors



ChatGPT



Twitter



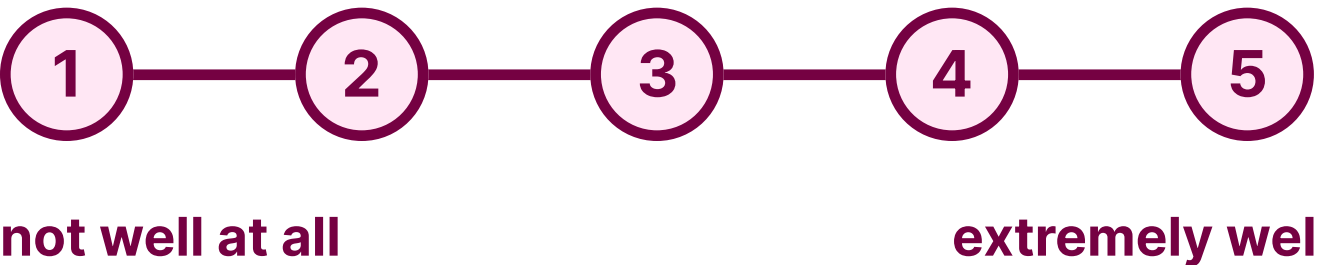
YouTube

Deploying a method for comparing design and user metaphors



Twitter is

because



identifying design metaphors

what metaphors do platforms project to users?



21 design metaphors

across all three platforms

eliciting user metaphors

what metaphors best describe users' experiences?



554 user metaphors

n=300 participants

rating user metaphors

how well does a user metaphor reflect others' experiences?



~50 ratings per metaphor

n=926 participants

**So, what do we learn from
comparing design and user
metaphors?**

First, we learn *which* design metaphors translate to users

For some design metaphors, we find both *explicit* matches...



ChatGPT is an **assistant** because it facilitates task completion

ChatGPT is an **personal assistant** because it assists you in coming up with new ideas for your work

ChatGPT is an **well-versed assistant** because it helps transform your thoughts into words

...and *related* matches among users' metaphors.

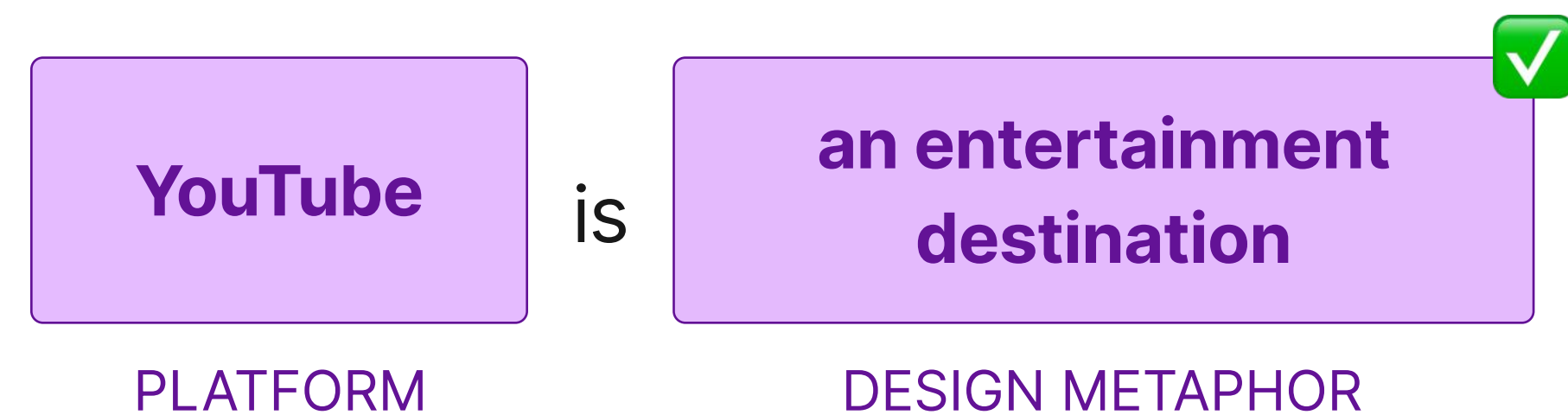


ChatGPT is a **librarian** because it helps you find answers to things you don't know

ChatGPT is a **receptionist** because you can ask them questions about something and you get answers

ChatGPT is a **helper** because it helps me when I can't find the information on Google

Other design metaphors do not exactly translate, but have *related* matches...



YouTube is a **circus** because it is entertaining and there is a wide variety of things to see.

YouTube is an **old video rental store** because I can get my fill of 80s movie reviews

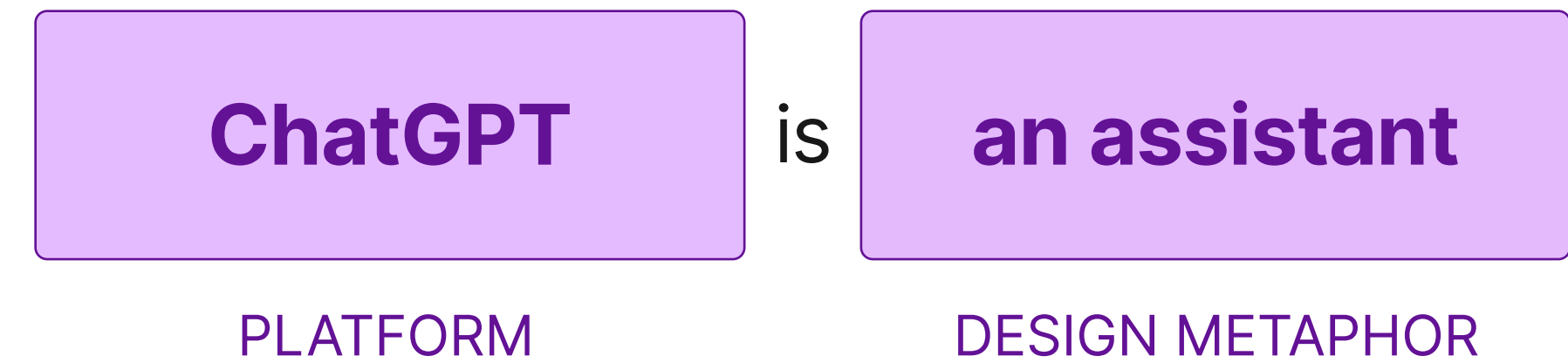
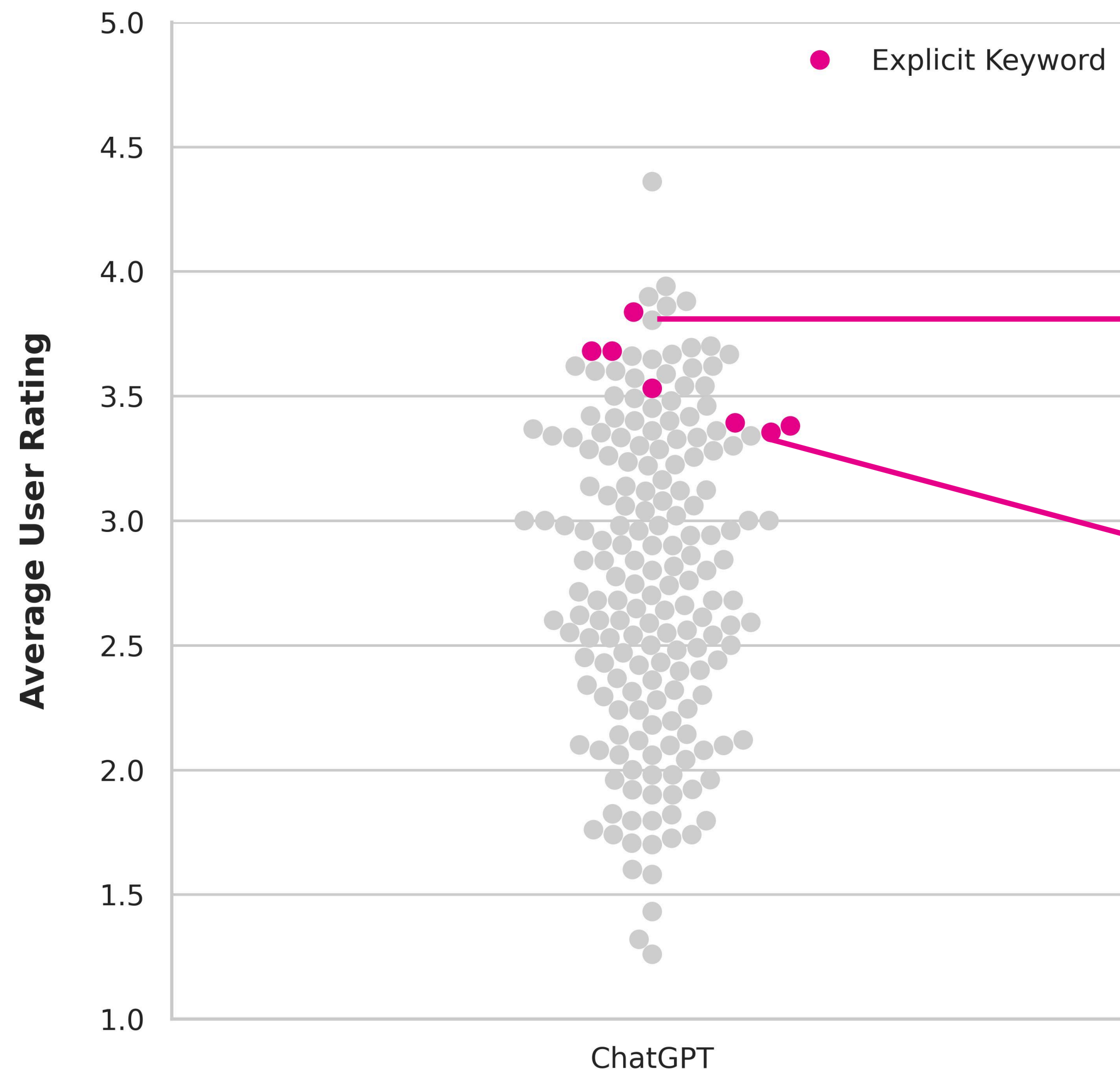
YouTube is an **exhibit** because it's a collection of many creative works from around the world

...and a few design metaphors *fail* to translate at all.



We also learn *how well* design metaphors reflect users' experiences.

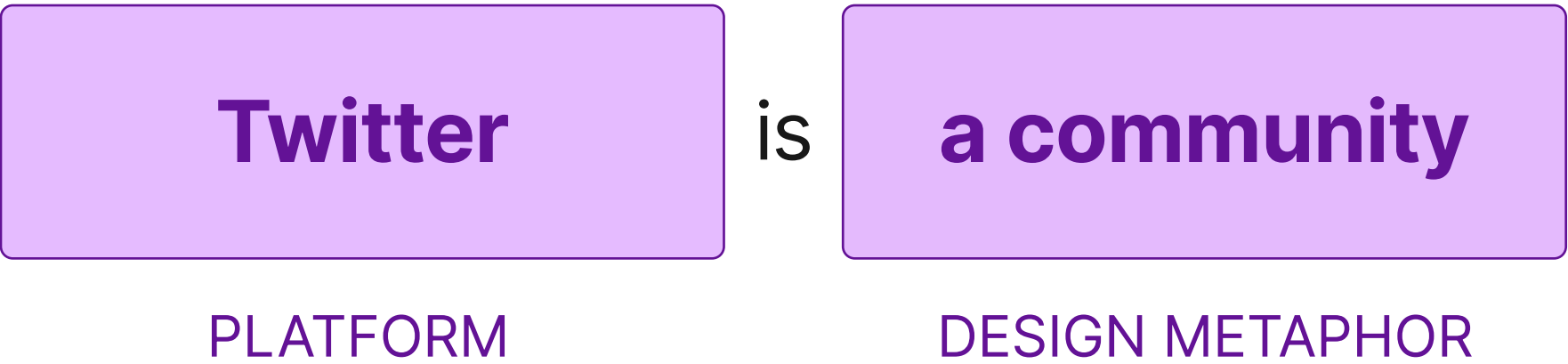
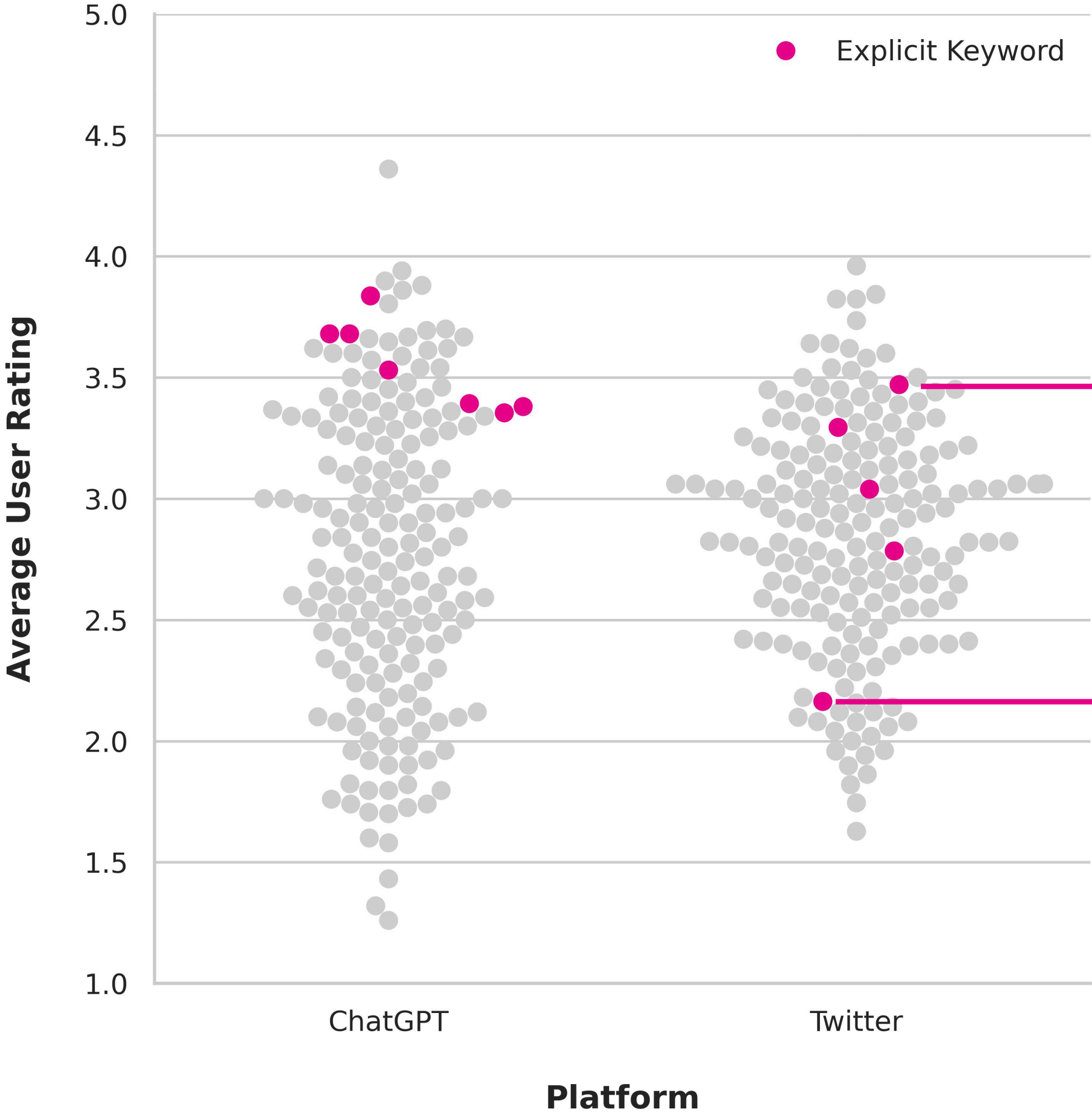
For some design metaphors, user metaphors that **explicitly** match are *highly* reflective.



ChatGPT is **an assistant** because it facilitates task completion

ChatGPT is **a well-versed assistant** because it helps transform your thoughts into words

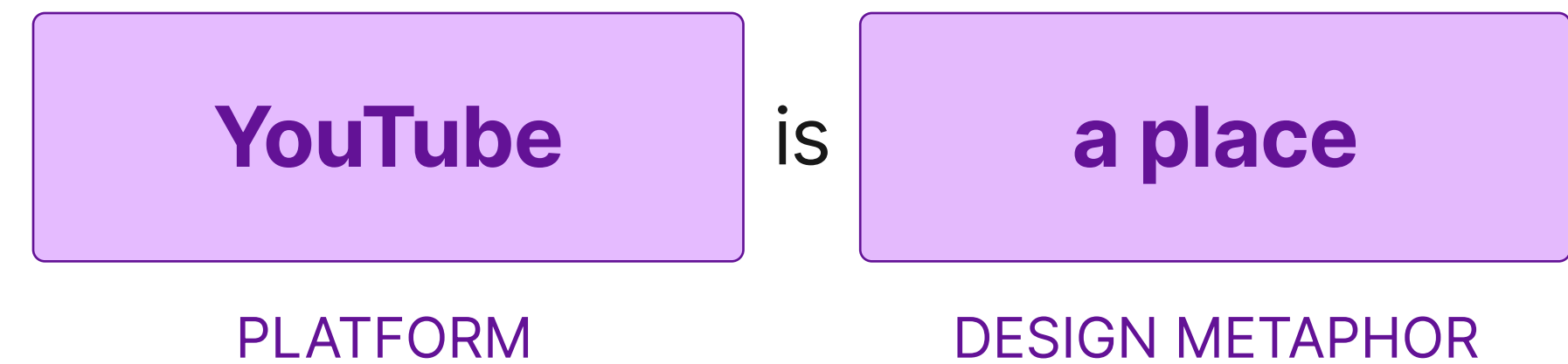
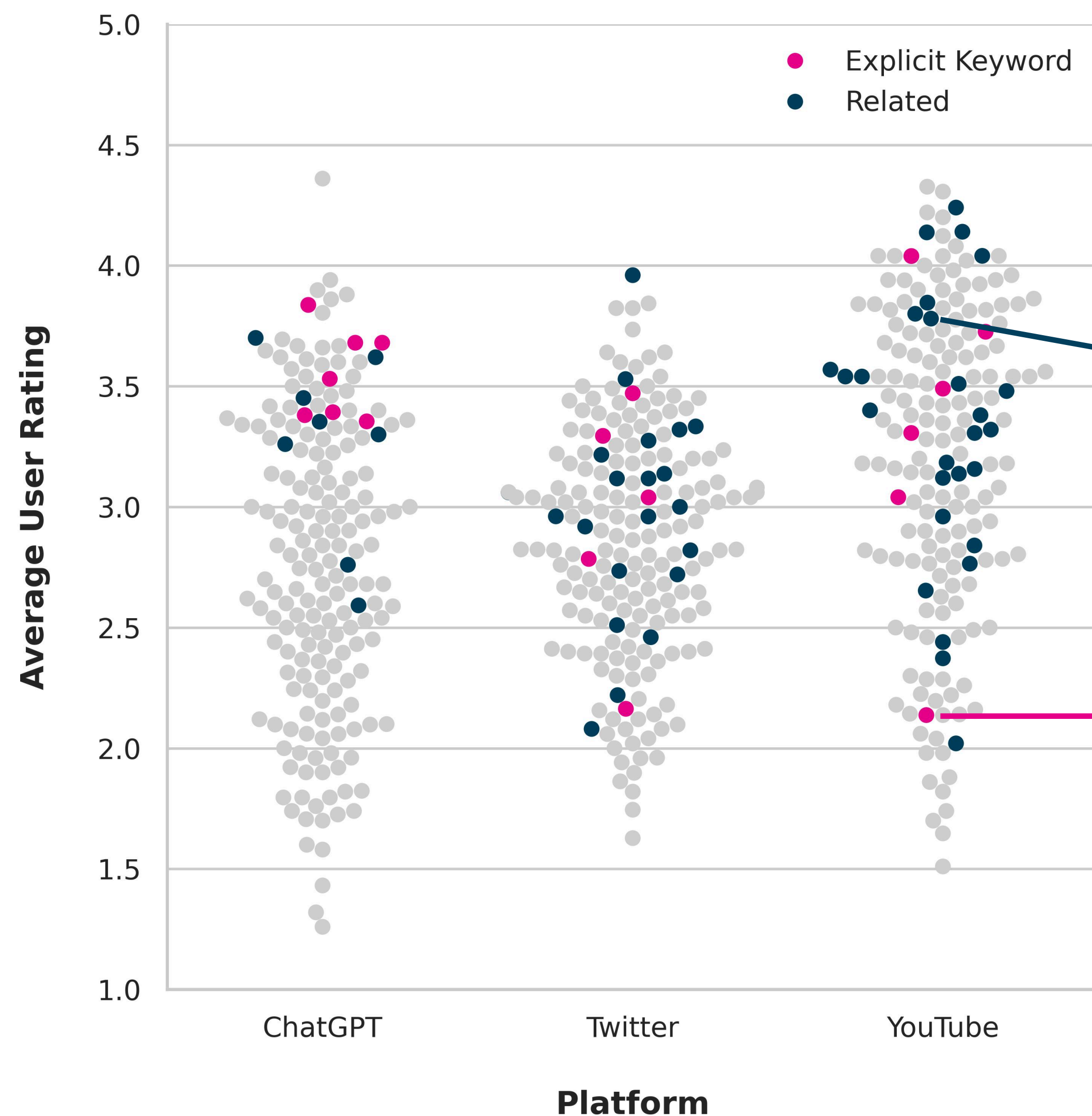
For others, user metaphors that **explicitly match** are *not always reflective* of users' experiences...



Twitter is **a community** because people can come together to share ideas

Twitter is a **more diverse thought community** because the censorship has stopped

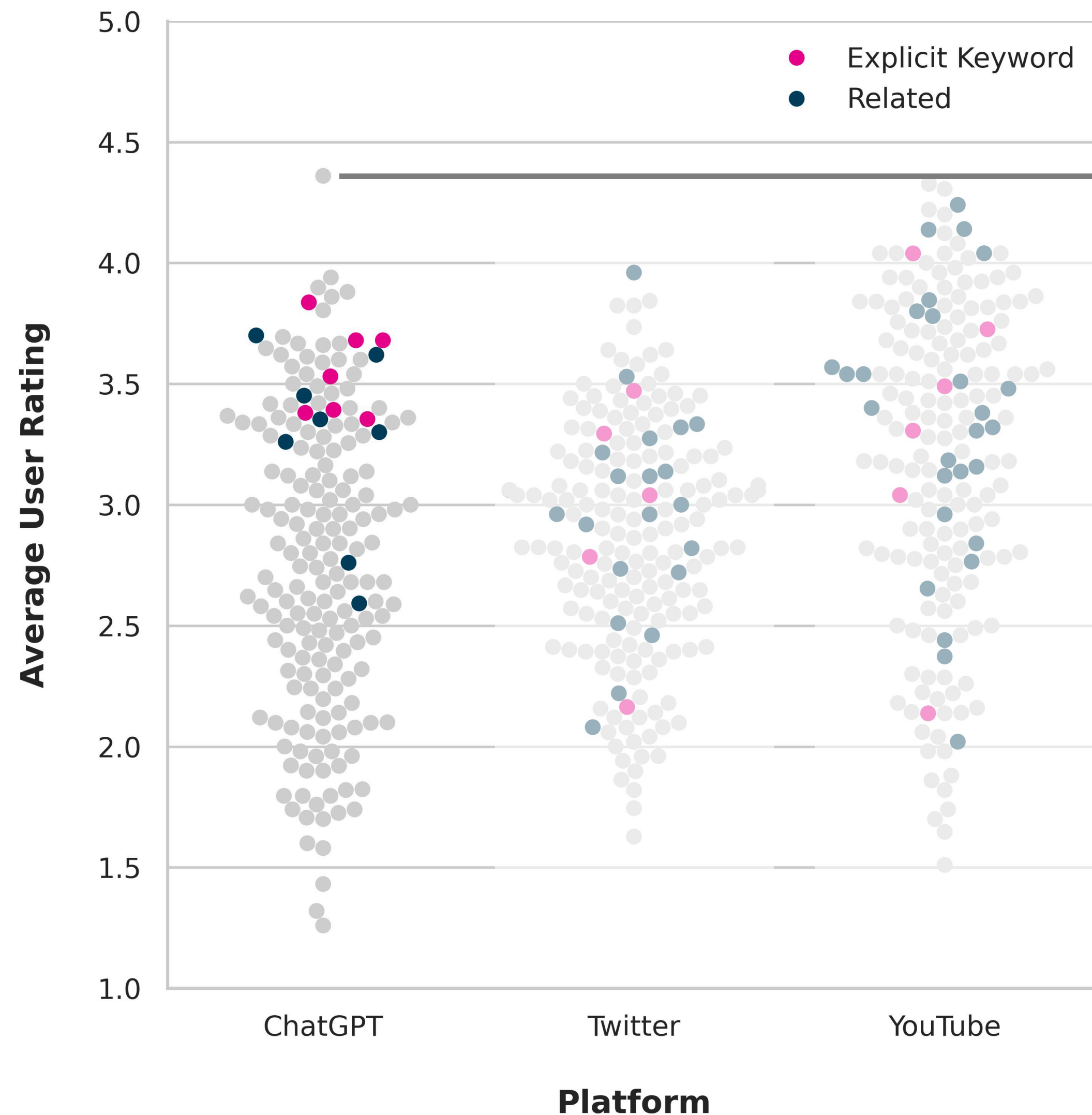
...or user metaphors that are different, but *related* are more *reflective*.



YouTube is **an open world** because you can see videos on different things

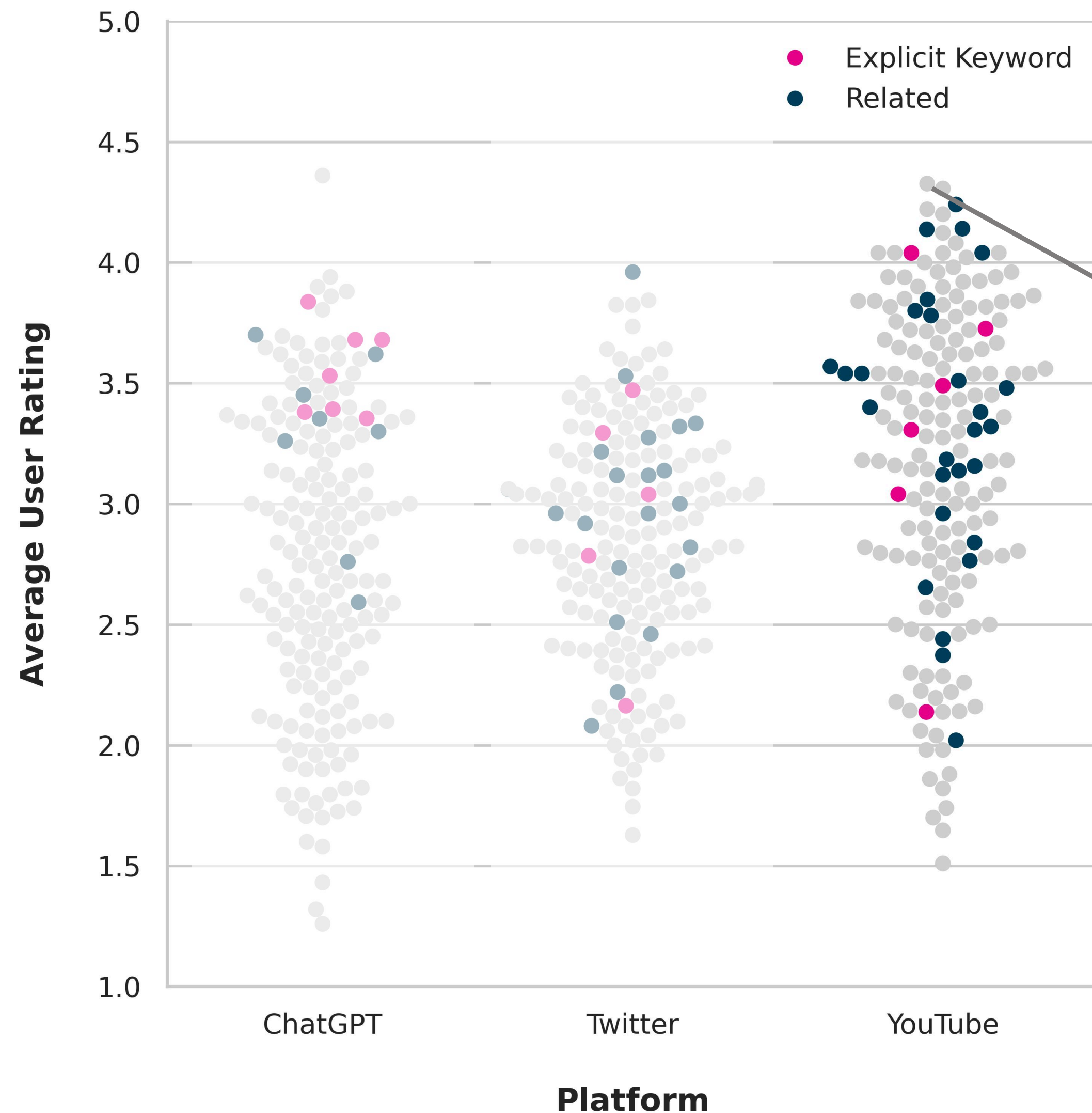
YouTube is **a gathering place** because you are able to meet and connect with loved ones

Yet, the highest rated metaphors *may not* explicitly match or even be related to design metaphors...



ChatGPT is **a tool** because you can use it for work

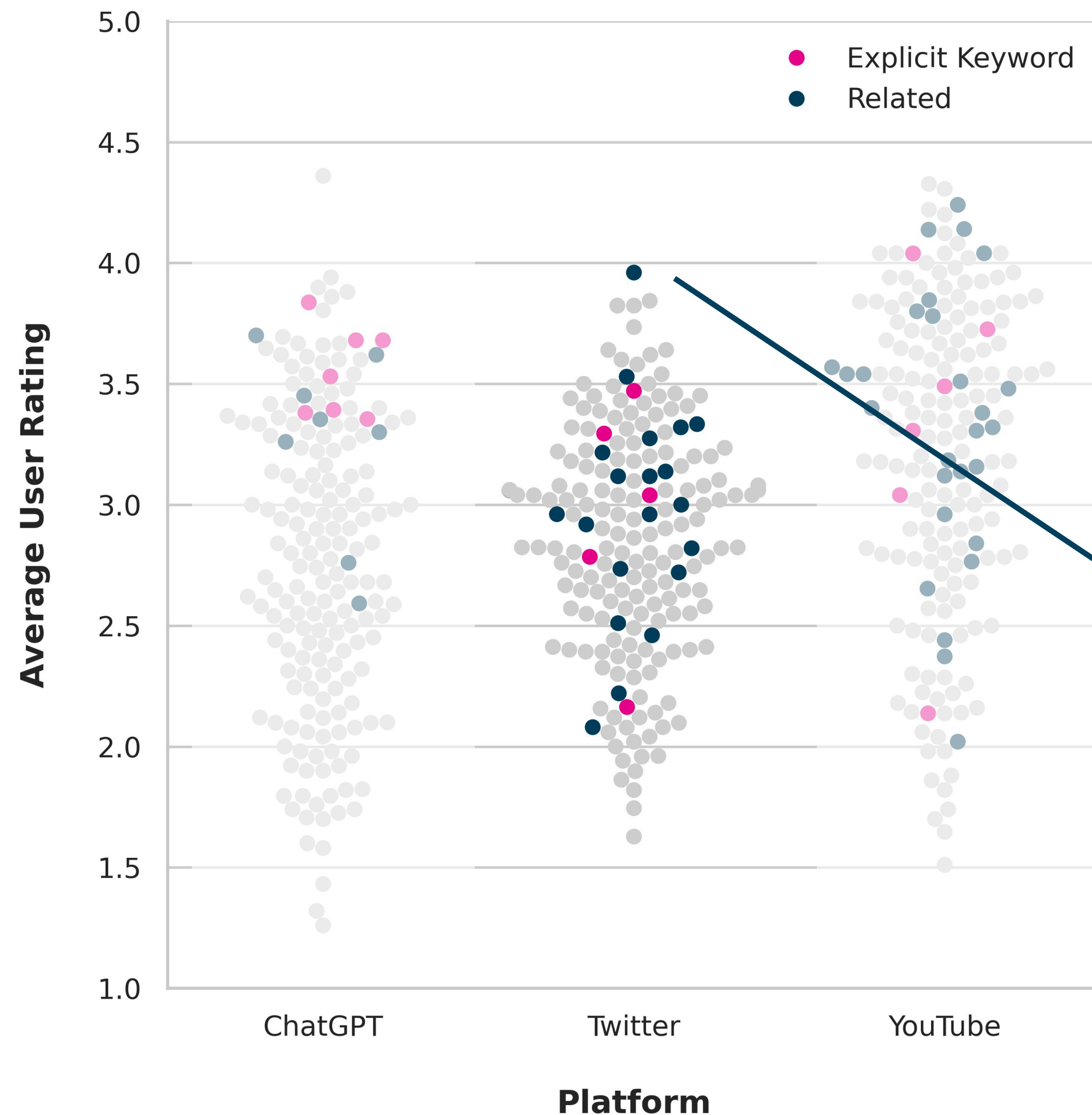
Yet, the highest rated metaphors *may not* explicitly match or even be related to design metaphors...



ChatGPT is **a tool** because you can use it for work

YouTube is **a library** because you are able to meet and connect with loved ones

Yet, the highest rated metaphors *may not* explicitly match or even be related to design metaphors...

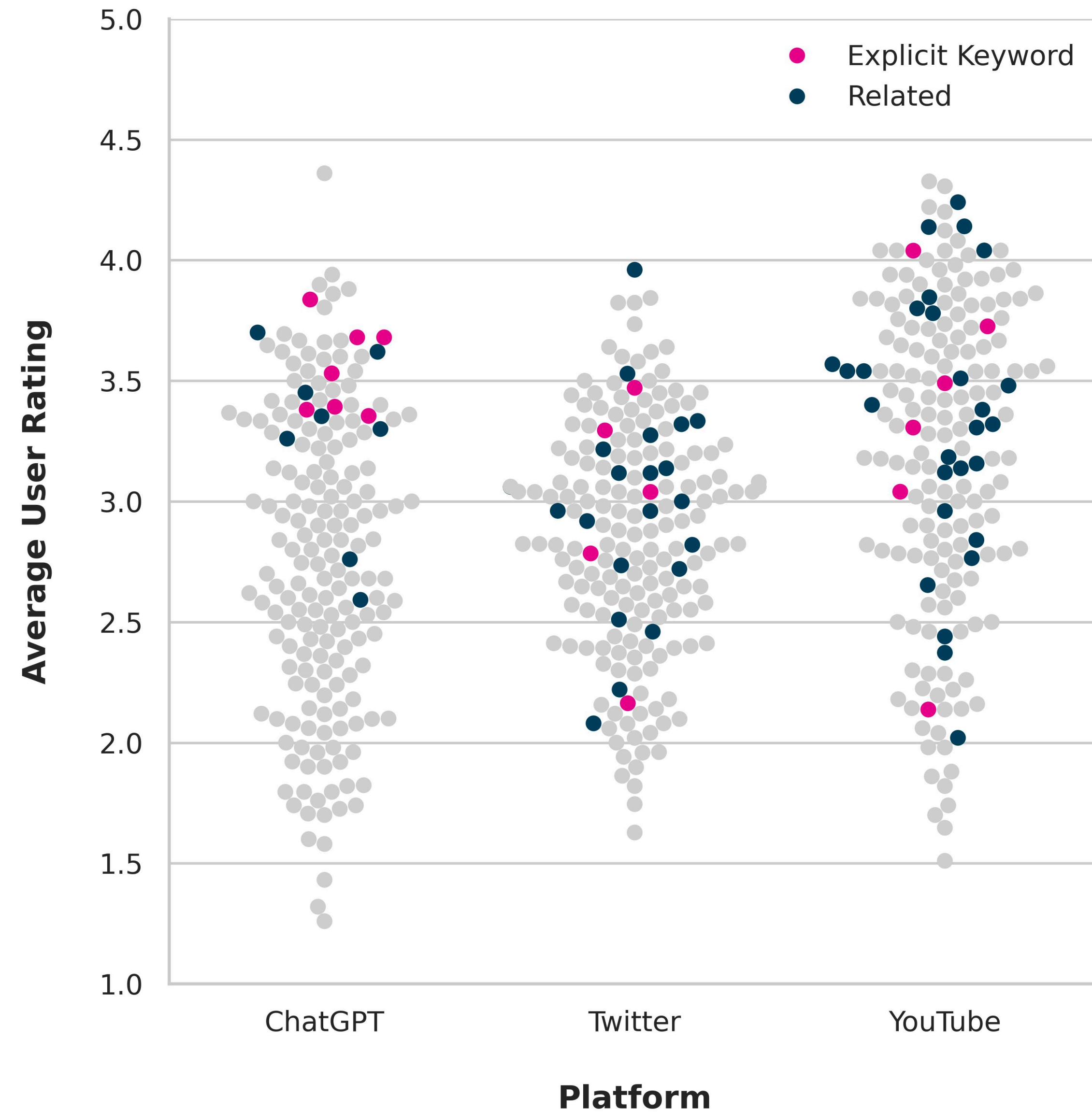


ChatGPT is **a tool** because you can use it for work

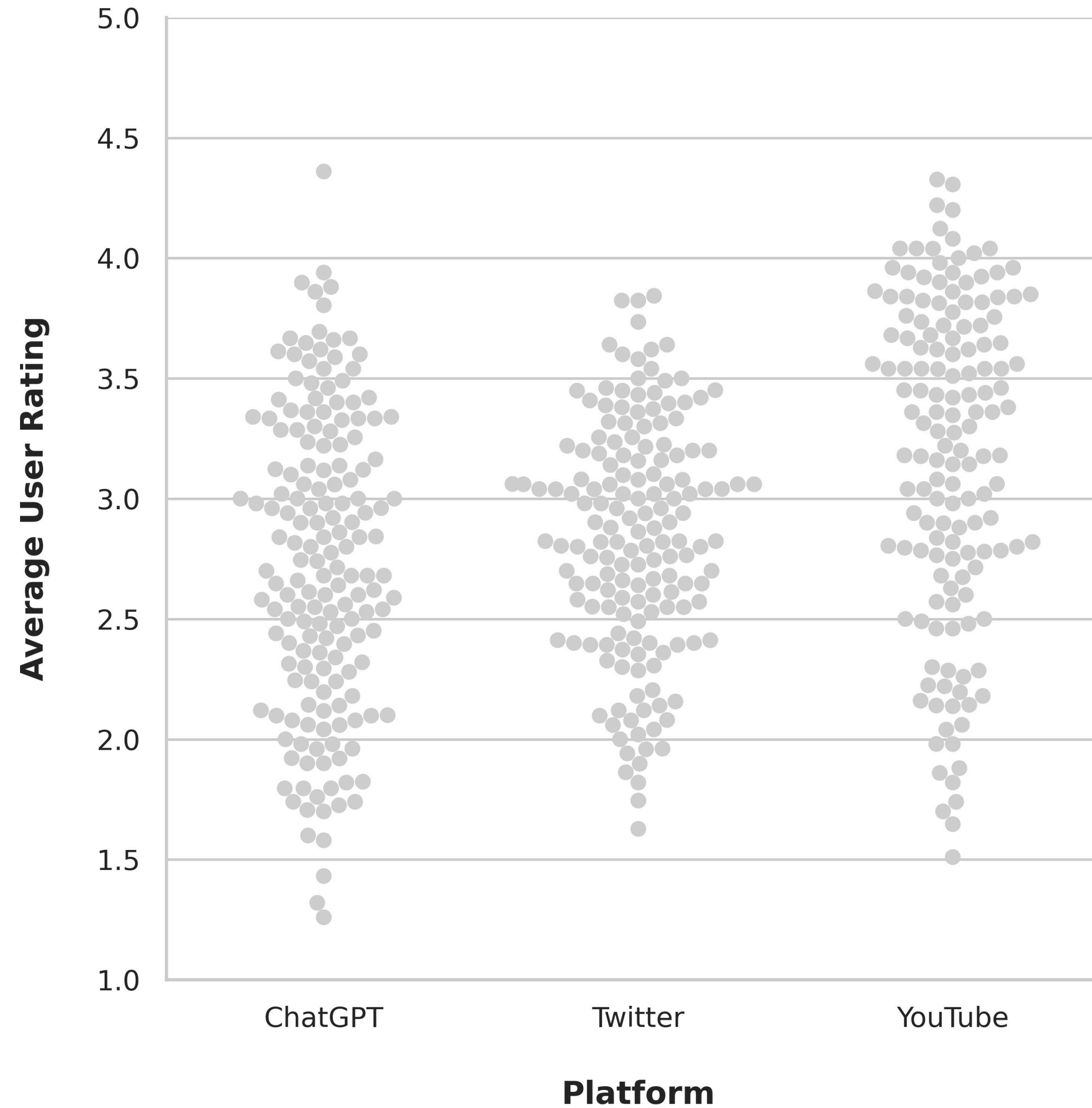
YouTube is **a library** because you are able to meet and connect with loved ones

Twitter is **a chaotic town square** because everyone can communicate their ideas but it's difficult to tell fact from fiction

...and most user metaphors do not match design metaphors.



...and most user metaphors do not match design metaphors.



Finally, we learn about user and community experiences that aren't captured by design metaphors.

There are many user metaphors beyond ones related to design metaphors...

ChatGPT is **a library help desk** because it has an answer for everything

ChatGPT is **a dollar scratch off ticket** because sometimes you win big but sometimes you strike out

ChatGPT is **a wishing well** because like a coin, you throw in a request and wish for the best

ChatGPT is **a Taco Bell meal** because its awesome right now but you're gonna pay for it later

ChatGPT is **an expanding mind** because it is constantly learning and trying to evolve into a better version of itself

ChatGPT is **a good improv partner** because it knows just how to respond

ChatGPT

is

an assistant

PLATFORM

DESIGN METAPHOR

...and user experiences beyond those encapsulated in design metaphors.



...and user experiences beyond those encapsulated in design metaphors.

creating communication

EXPERIENCE

versus

self-expression

USER EXPERIENCE

Twitter is **a town hall** because everyone shares their opinions

Twitter is **a free space** because anyone can say anything

Twitter is **a canvas of thoughts** because you can express whatever comes into you mind in writing

...and user experiences beyond those encapsulated in design metaphors.

creating communication

EXPERIENCE

versus

toxicity

USER EXPERIENCE

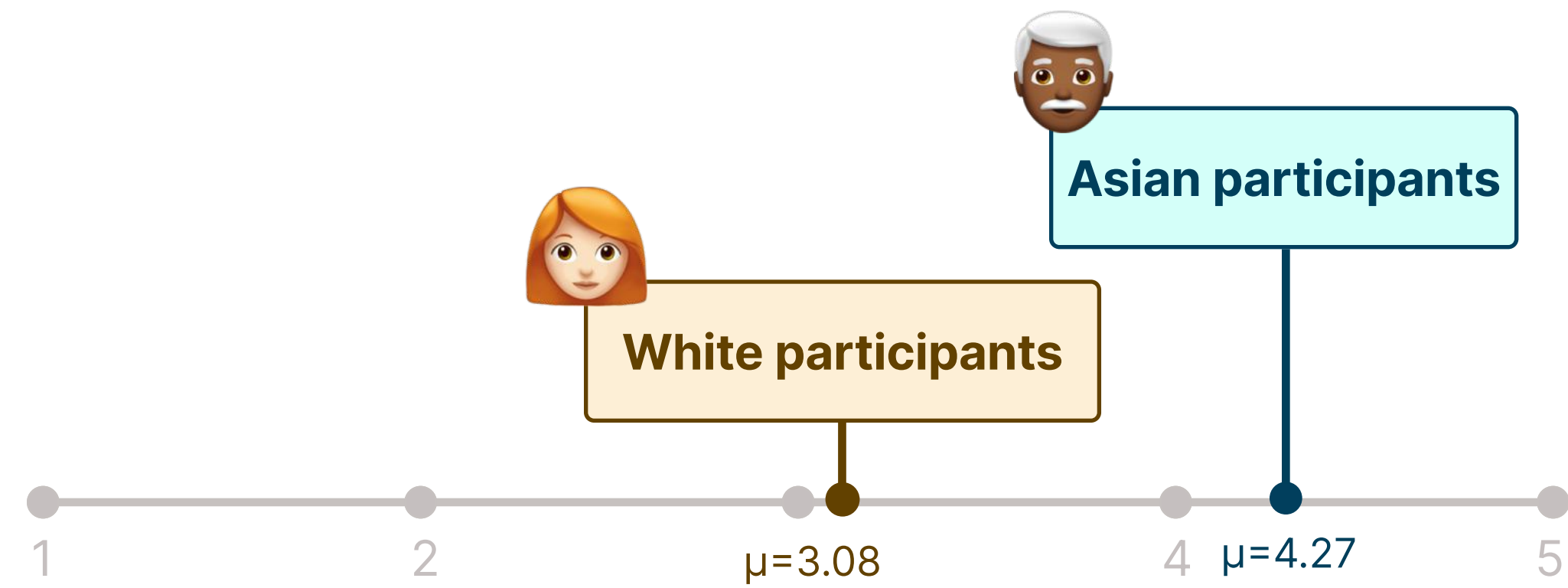
Twitter is **a swamp** because it often breeds unkindness

Twitter is **a toxic waste plant** because there is so much negative energy and toxic comments

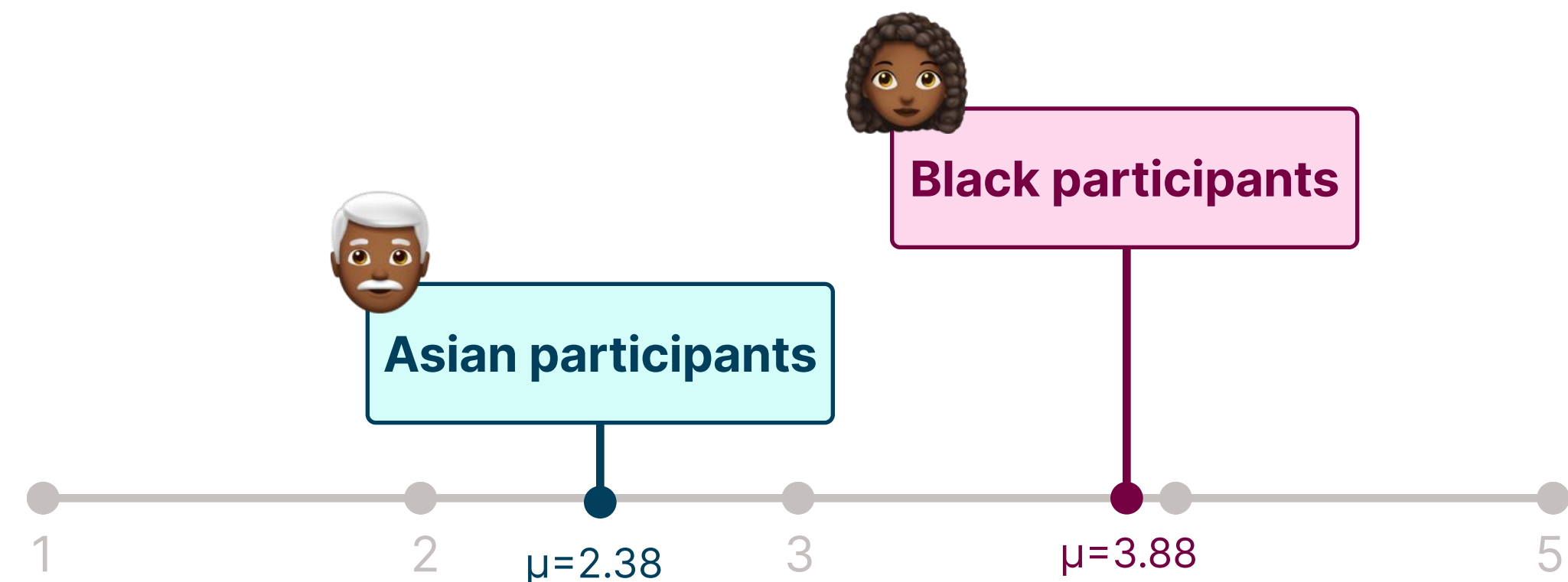
Twitter is **high school** because it's filled with drama

We also surface metaphors that reflect the experiences of specific communities more than others.

Twitter is **a maze** because there are so many false paths before getting the information you want.

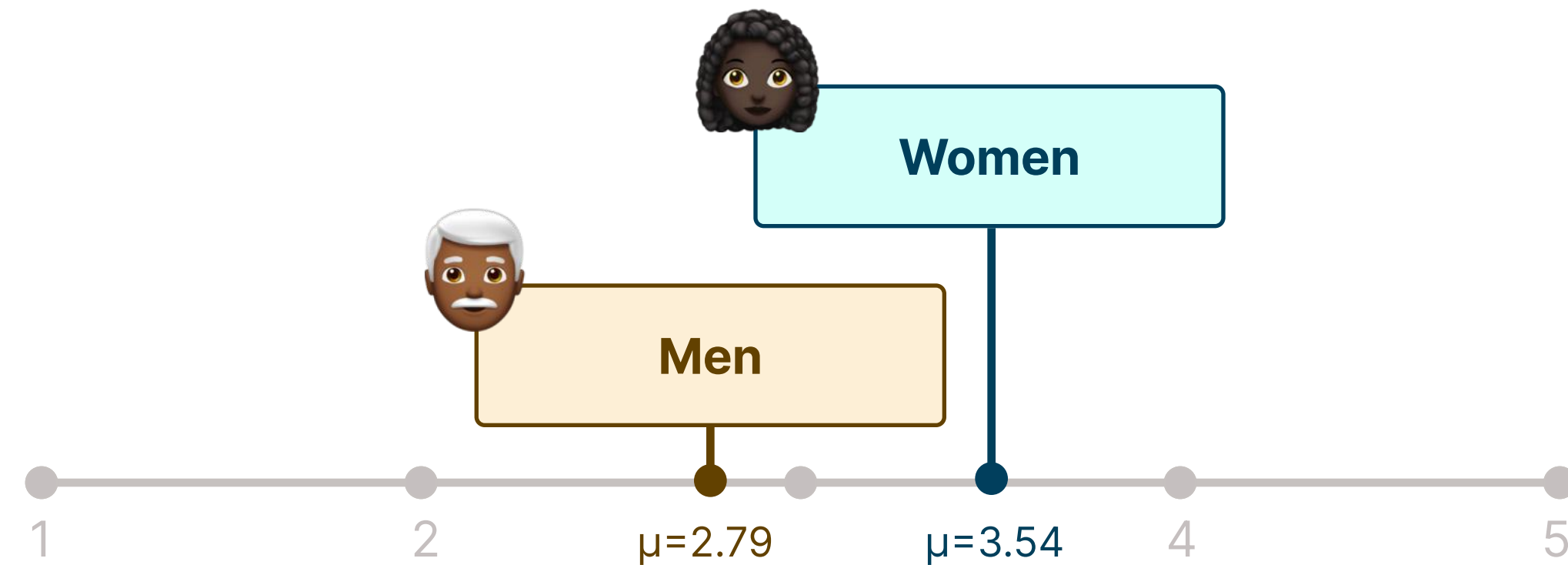


Twitter is **a diary** because many share their inner most thoughts

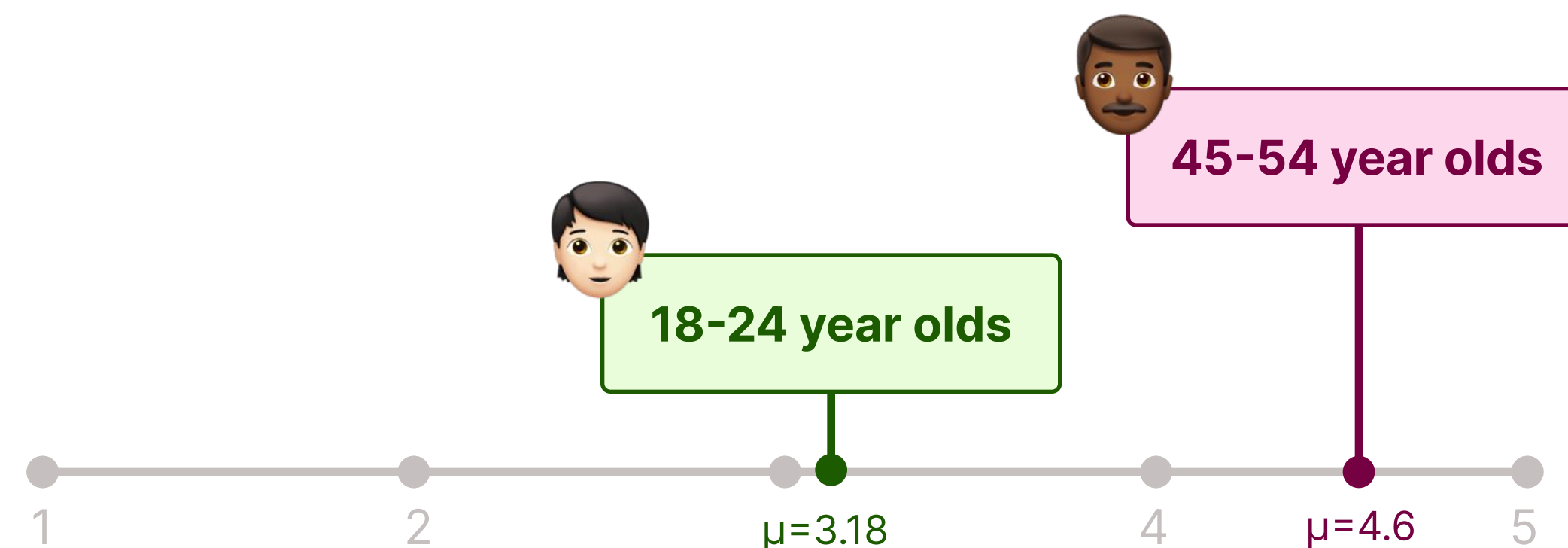


We also surface metaphors that reflect the experiences of specific communities more than others.

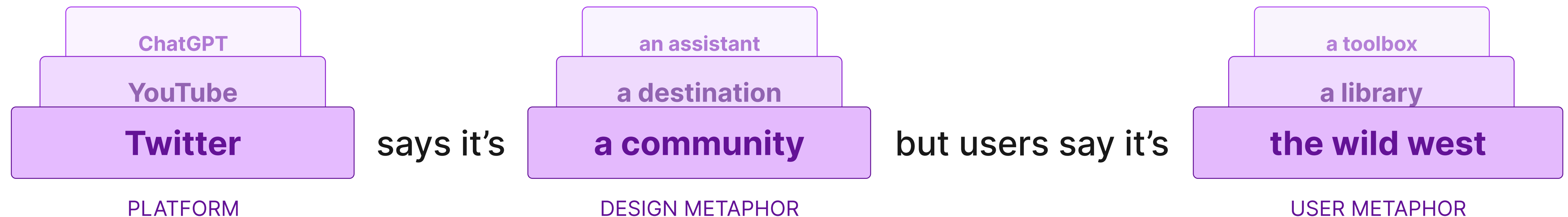
Twitter is **a diary** because many share their inner most thoughts



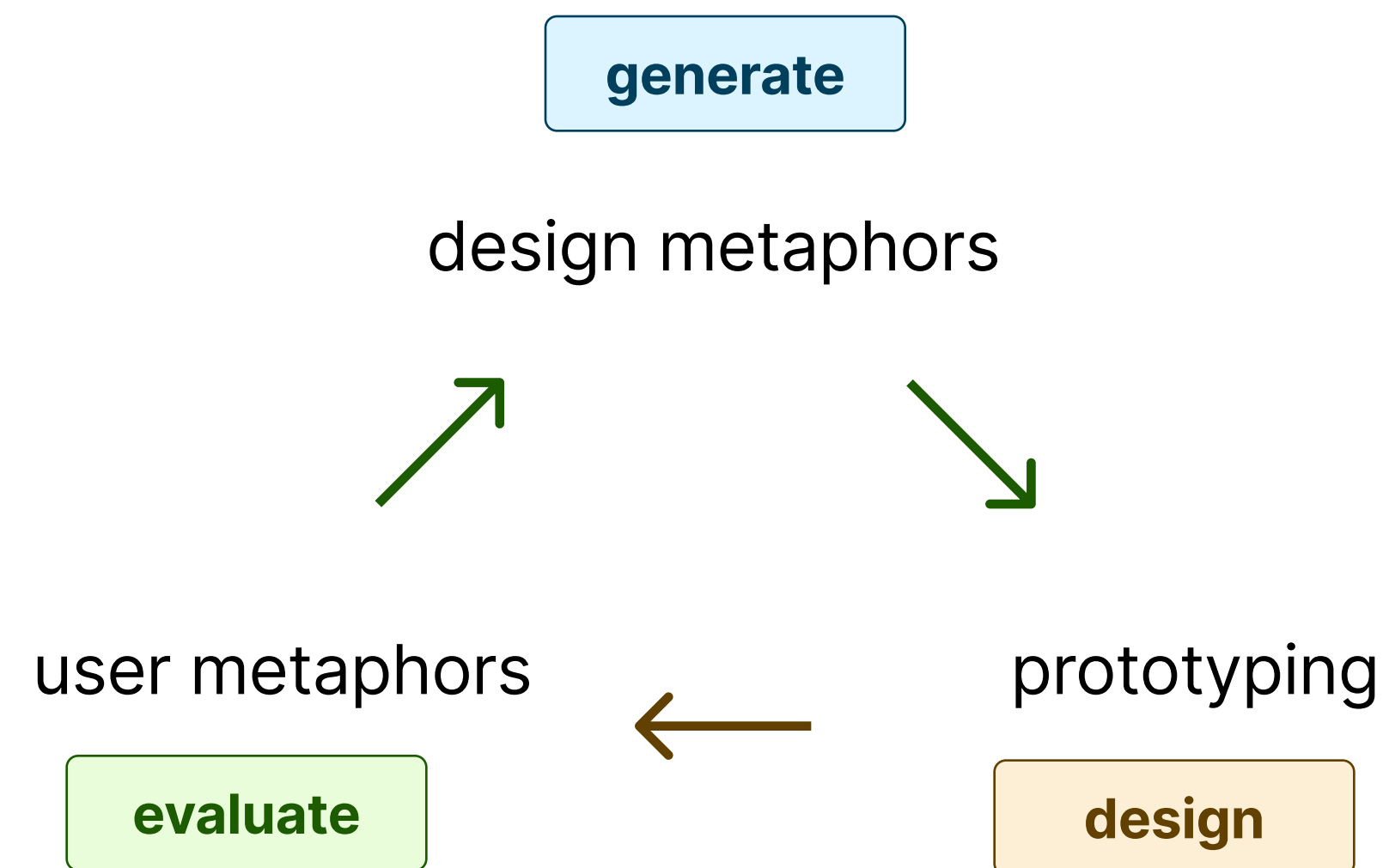
Twitter is **kindergarten** because it is full of immature arguments



Through comparison, we learn *if, how well, and for whom* design metaphors translate.



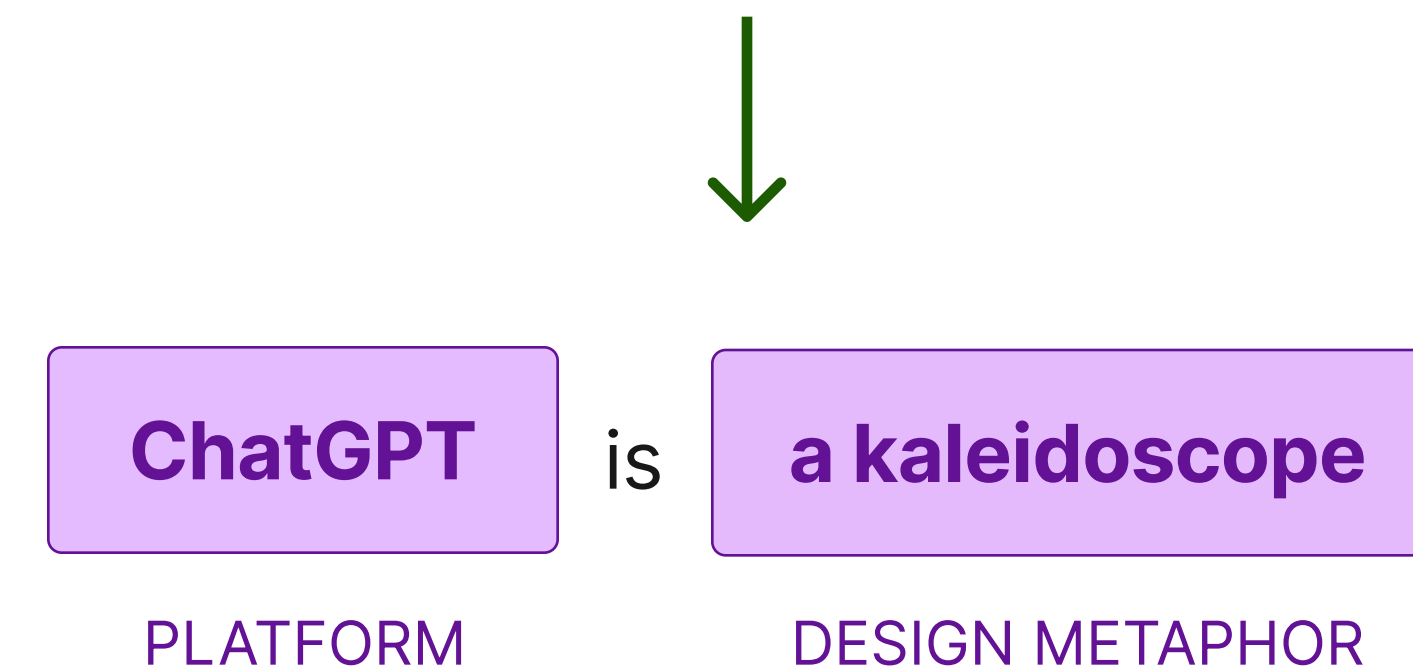
Supporting an iterative process around design metaphors



feedback for effective design metaphors

evaluating design metaphors in prototypes and post-initial development

ChatGPT is **a kaleidoscope** because it offers a constantly shifting array of perspectives and insights



generating new ideas for design metaphors

generating new design metaphors via unique user metaphors and experiences

Comparing Design Metaphors and User-Driven Metaphors



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beleicia@cs.stanford.edu